Microbiology Labs

Alveary

Microbiology Labs

GRADE(S): 12

About the Course

Labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.

This topic is included in the following course(s): Science: Microbiology

Scheduling

GRADE	SCHEDULE INFO.	BOOKS
12	Microbiology Labs 1 time/week 60 min	Illustrated Guide to Home Biology Experiments

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Microbiology				
Microbiology Lessons Nature Walks & Scouting: Grades 9-12	Microbiology Lessons	Microbiology Lessons	Microbiology Lessons Nature Notebook: Grades 9-12	Microbiology Lessons Microbiology Labs

Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details]

Microbiology Labs

Illustrated Guide to Home Biology Experiments

Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ View Basic Supplies

(No Subject Supplies Assigned)

Quick Links

(No Quick Links Assigned)

Click THIS text or scan the QR code for links.

Microbiology Labs

How To Approach Labs

Introduce

- Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.

- Relevant concept(s) are introduced in the text.

- Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.

Lab Procedure

- Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.

- Record all data and observations in the lab notebook.

Analysis & Conclusions

- After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.

- If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.

Microbiology Labs

[Click THIS text or scan the QR code for links.]

Term 1

WEEK 1 ⬚ **60m Microbiology Labs - Lesson 1**

Materials: Illustrated Guide to Home Biology Experiments

➔ LAB DAY
Continue reading p.3-51. Then, skim through the rest of the book and plot out a rough plan of which experiments you'd like to explore. Order the needed supplies. Home Scientist has a kit that contains the basic supplies needed. You can add on bacterial cultures and prepared slides, if desired. If you order supplies elsewhere, you can still order the DVD with answers to the review questions in the book and high-resolution images as a separate item. As the introductory material in the book says, if you decide to do labs that require bacterial cultures or prepared slides of certain items that you cannot procure locally, you will need to order those items from Home Scientist ahead of time. They suggest that you could use the high-resolution images that come with their DVD kit instead of purchasing slides.

➔ VIEW
∞ Link: Home Scientist

WEEK 2 ⬚ **60m Microbiology Labs - Lesson 2**

Materials: Illustrated Guide to Home Biology Experiments

➔ LAB DAY
Today, you will complete LAB I-1: Using a Microscope. Gather all of the needed materials. Prepare your Lab Notebook for the experiment. Read the lab carefully before beginning and adhere to all safety requirements. Record what you do and what you observe in your Lab Notebook. Then clean up and dispose of any materials as instructed.

Complete the lab write-up in your notebook, including the review questions. After you have finished your work, use the answers in the PDF that came with your supplies and see if there was anything significant that you did not understand. Use your Biology textbook from Grade 10 and the internet to learn more or aid in understanding terms and concepts.

You will complete your lab report following the book's guidelines. Choose 6 throughout the year to complete as full lab reports. Use what you have learned about writing lab reports. Use the guidelines for the rest of your reports.

WEEK 3 ⬚ **60m Microbiology Labs - Lesson 3**

Materials: Illustrated Guide to Home Biology Experiments

➔ LAB DAY